

ECON 712: Applied Econometrics

Lecture 11: Statistical Properties of OLS, part I

CEF and Linear Regression

Finite Sample Properties OLS

Asymptotic Properties OLS

Why Linear Regression?

- Suppose that the population CEF is linear and has the following functional form:

$$\mathbb{E}[y_i|\mathbf{X}_i] = \mathbf{X}_i^T \beta^* \text{ (MHE notation)}$$

- ▶ by the CEF Decomposition Theorem:

$$\mathbb{E}[\mathbf{X}_i(y_i - \mathbb{E}[y_i|\mathbf{X}_i])] = \mathbf{0}$$

- ▶ Substituting for the linear CEF:

$$\mathbb{E}[\mathbf{X}_i(y_i - \mathbf{X}_i^T \beta^*)] = \mathbf{0}$$

- ▶ Solving for $\beta^* = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T]^{-1} \mathbb{E}[\mathbf{X}_i y_i] = \beta^{OLS}$

- Interpretation: OLS solves the same problem as the linear CEF

Why Linear Regression?

- What if the CEF is **not linear**?

1. By definition, β^{OLS} provides the best linear prediction of y_i :

$$\beta^{OLS} = \underset{b}{\operatorname{argmin}} \mathbb{E}[(y_i - \mathbf{X}_i^T \mathbf{b})^2]$$

2. β^{OLS} is the best linear approximation to the CEF:

$$\beta = \underset{b}{\operatorname{argmin}} \mathbb{E}[(\mathbb{E}[y_i | \mathbf{X}_i] - \mathbf{X}_i^T \mathbf{b})^2]$$

- Proof: by FOC

$$\mathbb{E}[\mathbf{X}_i(\mathbb{E}[y_i | \mathbf{X}_i] - \mathbf{X}_i^T \beta)] = \mathbf{0} \Rightarrow$$

$$\beta = \mathbb{E}[\mathbf{X}_i \mathbf{X}_i^T]^{-1} \mathbb{E}[\mathbf{X}_i y_i] = \beta^{OLS}$$

where the last line uses the CEF decomposition property.

Linear Regression and CEF



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Unbiasedness

- Under strict exogeneity, $\mathbb{E}[\mathbf{u}|\mathbf{X}] = \mathbf{0}$, OLS/MM is unbiased:

$$\hat{\beta}^{ols} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$$

$$\hat{\beta}^{ols} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$$

$$\mathbb{E}[\hat{\beta}^{ols}|\mathbf{X}] = \beta \underbrace{=}_{\text{by LIE}} \mathbb{E}[\hat{\beta}^{ols}]$$

- Pre-determinedness, $\mathbb{E}[u_i|x_{1i}, \cdot, x_{ki}] = 0$, is not enough
- The linear CEF has to be correctly specified

Variance under Spherical Errors

- Another important property of OLS is the conditional **variance-covariance** matrix of estimates:

$$\text{Var}[\hat{\beta}^{ols}|\mathbf{X}] = \mathbb{E}[(\hat{\beta}^{ols} - \beta)(\hat{\beta}^{ols} - \beta)^T|\mathbf{X}]$$

- Assume that $\mathbb{E}[\mathbf{u}\mathbf{u}^T|\mathbf{X}] = \sigma^2 \mathbf{I}$, then

$$\text{Var}[\hat{\beta}^{ols}|\mathbf{X}] = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}$$

- ▶ In practice, we estimate σ^2 by $\hat{\sigma}^2 = \hat{\mathbf{u}}^T \hat{\mathbf{u}}(n - k)^{-1}$
- Spherical error terms is a very strict assumption, which is not usually satisfied, specially if the CEF is not linear
- By the law of total variance, the unconditional variance is

$$\text{Var}(\hat{\beta}^{ols}) = \sigma^2 \mathbb{E}[(\mathbf{X}^T \mathbf{X})^{-1}]$$

Distribution of regressors under Gauss-Markov

- Classical or Gauss Markov Assumptions of Multiple Regression are:
 1. If the population equation is given by (linear CEF)

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$$

2. No perfect multicollinearity (i.e. $\mathbf{X}$ has full rank)
 3. Zero Conditional Mean (i.e. $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$)
 4. Homoskedasticity and no serial correlation (i.e. $\text{Var}[\mathbf{u}|\mathbf{X}] = \sigma^2 \mathbf{I}_n$)
 5. Population conditional error terms are jointly Normal:
 $\mathbf{u}|\mathbf{X} \sim N[\mathbf{0}, \sigma^2 \mathbf{I}_N]$
- This yield the following result:

$$\hat{\beta}|\mathbf{X} \sim N[\beta, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}]$$

- Because of properties of the multivariate normal distribution, this yields the following marginal distribution of estimators:

$$\hat{\beta}_j|\mathbf{X} \sim N[\beta_j, \sigma^2 (\mathbf{X}^T \mathbf{X})_{jj}^{-1}]$$

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Consistency

- Classical Assumptions are very restrictive and not usually satisfied in practice
- Therefore, modern econometrics relies on asymptotic properties of estimators, which are satisfied under weaker conditions
- For instance, if we assume that $\mathbb{E}[\mathbf{X}_i^T u_i] = \mathbf{0}$, then

$$\begin{aligned} \underset{n \rightarrow \infty}{plim} \hat{\beta}^{ols} &= \underset{n \rightarrow \infty}{plim} \left[\beta + \left(\frac{1}{n} \mathbf{X}^T \mathbf{X} \right)^{-1} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right) \right] \\ &= \beta + \left(\mathbf{S}_{\mathbf{X}^T \mathbf{X}} \right)^{-1} \underset{n \rightarrow \infty}{plim} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right) \\ &= \beta \text{ (the OLS estimator is consistent)} \end{aligned}$$

- Because we do not need mean independence, but zero covariance, the CEF does not need to be linear
- Pre-determinedness is enough